

Chapter 02: First-Order Differential Equations & Modeling

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Course: Applied Ordinary Differential Equations (ENGR 213) | **Textbook:** Dennis G. Zill (7th Ed., Chapter 2)

1. Executive Summary & First-Principles Philosophy

Chapter 2 forms the analytical and practical backbone of first-order differential equations. In this chapter, we master the 5 core analytical solution techniques:

1. **Separation of Variables** ($g(y)dy = f(x)dx$)
 2. **Integrating Factors for Linear Equations** ($y' + P(x)y = Q(x)$)
 3. **Exact Equations & Multipliers** ($Mdx + Ndy = 0$)
 4. **Substitutions** (Homogeneous, Bernoulli, Linear Composition)
 5. **Applied Physical Modeling** (Newton's cooling, brine tanks, logistic population dynamics)
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2. Core Mechanics & Mathematical Engine

A. Separation of Variables

If an ODE can be factored as $\frac{dy}{dx} = g(x)h(y)$, divide by $h(y)$ (noting any zero-division singular solutions where $h(y) = 0$) and integrate both sides:

$$\int \frac{1}{h(y)} dy = \int g(x) dx + C$$

B. First-Order Linear ODEs & The Integrating Factor

Standard linear form requires the leading coefficient of $\frac{dy}{dx}$ to be strictly 1:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Multiply every term by the integrating factor $\mu(x) = e^{\int P(x)dx}$:

$$e^{\int P(x)dx} \frac{dy}{dx} + P(x)e^{\int P(x)dx} y = Q(x)e^{\int P(x)dx}$$

Recognize the left side as the product rule: $\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$. Integrate and divide:

$$y(x) = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x)dx + C \right]$$

C. Exact Equations & Integrating Multipliers

An equation $M(x, y)dx + N(x, y)dy = 0$ is **exact** if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Then there exists a potential function $F(x, y)$ such that $\frac{\partial F}{\partial x} = M$ and $\frac{\partial F}{\partial y} = N$. The general solution is $F(x, y) = C$.

If not exact, calculate:

1. If $\frac{M_y - N_x}{N}$ depends only on x , then $\mu(x) = \exp\left(\int \frac{M_y - N_x}{N} dx\right)$.
2. If $\frac{N_x - M_y}{M}$ depends only on y , then $\mu(y) = \exp\left(\int \frac{N_x - M_y}{M} dy\right)$.

D. Substitution Archetypes

1. **Homogeneous Equations:** If $M(tx, ty) = t^n M(x, y)$ and $N(tx, ty) = t^n N(x, y)$, substitute $y = ux \implies dy = udx + xdu$.
2. **Bernoulli Equation:**

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Divide by y^n : $y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x)$. Substitute $w = y^{1-n}$, which transforms into linear:

$$\frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x)$$

3. **Linear Composition:** $\frac{dy}{dx} = f(Ax + By + C) \implies u = Ax + By + C \implies \frac{du}{dx} = A + B \frac{dy}{dx}$.
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3. Curriculum-Grounded Visual Reference

consequently it may be possible to see at a glance certain qualitative aspects of the solutions—regions in the plane, for example, in which a solution exhibits an unusual behavior. A single solution curve that passes through a direction field must follow the flow pattern of the field; it is tangent to a lineal element when it intersects a point in the grid.

EXAMPLE 1 Direction Field

The direction field for the differential equation $dy/dx = 0.2xy$ shown in **FIGURE 2.1.2(a)** was obtained using computer software in which a 5×5 grid of points (mh, nh) , m and n integers, was defined by letting $-5 \leq m \leq 5$, $-5 \leq n \leq 5$ and $h = 1$. Notice in Figure 2.1.2(a) that at any point along the x -axis ($y = 0$) and the y -axis ($x = 0$) the slopes

Figure 2.1: Direction field and solution curves — from Zill 7th Ed. Chapter 2 (Fig. 2.1.2).

use the fact that $\frac{1}{4}A_0 = A(5730)$ or $\frac{1}{4}A_0 = A_0e^{5730k}$. The last equation implies $5730k = \ln \frac{1}{4} = -\ln 2$ and so we get $k = -(\ln 2)/5730 = -0.00012097$. Therefore $A(t) = A_0e^{-0.00012097t}$. With $A(t) = 0.001A_0$ we have $0.001A_0 = A_0e^{-0.00012097t}$ and $-0.00012097t = \ln(0.001) = -\ln 1000$. Thus

$$t = \frac{\ln 2}{0.00002867} \approx 24,180 \text{ years.} \quad \equiv$$

The date found in Example 3 is really at the border of accuracy of this method. The usual carbon-14 technique is limited to about 10 half-lives of the isotope, or roughly 60,000 years. One fundamental reason for this limitation is the relatively short half-life of C-14. There are other problems as well; the chemical analysis needed to obtain an accurate measurement of the remaining C-14 becomes somewhat formidable around the point 0.001A₀.

Figure 2.2: Mixing tank schematic: Input brine rate vs output drain rate — from Zill 7th Ed. Chapter 2 (Fig. 2.7.4).

4. Fully Worked Exam Archetype: The Draining Mixture Tank

Problem: A tank initially contains $V_0 = 300$ gal of brine in which $A_0 = 50$ lb of salt is dissolved. Pure water flows in at $r_{\text{in}} = 3$ gal/min, and the thoroughly mixed solution pumps out at $r_{\text{out}} = 2$ gal/min. Find the salt content $A(t)$ at any time t .

Step-by-Step Solution:

- **Step 1: Governing Mass Balance:**

$$\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}} = (c_{\text{in}}r_{\text{in}}) - (c_{\text{out}}(t)r_{\text{out}})$$

Since pure water enters, $c_{\text{in}} = 0 \implies R_{\text{in}} = 0$.

- **Step 2: Volume Formulation:**

$$V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t = 300 + (3 - 2)t = 300 + t \text{ gal}$$

Therefore, concentration at time t is $c_{\text{out}}(t) = \frac{A(t)}{300+t}$.

- **Step 3: Setup First-Order Linear ODE:**

$$\frac{dA}{dt} = -\frac{2A}{300+t} \implies \frac{dA}{dt} + \frac{2}{300+t}A = 0$$

- **Step 4: Separation / Integrating Factor:**

$$\frac{dA}{A} = -\frac{2}{300+t}dt \implies \ln|A| = -2\ln(300+t) + C_1$$

$$A(t) = \frac{C}{(300+t)^2}$$

- **Step 5: Apply Initial Condition:**

$$A(0) = 50 \implies 50 = \frac{C}{300^2} \implies C = 50 \times 90000 = 4,500,000$$

$$A(t) = \frac{4,500,000}{(300+t)^2} \text{ lb}$$

5. Rapid Exam Traps & Red Flags

- **⚠ Trap 1: Dropping $+C$ Before Exponentiating:** In separable equations, write $+C$ immediately after integrating. $e^{\ln|y|} = e^{kt+C} = C_0e^{kt}$. Never add $+C$ at the very end as an afterthought!
- **⚠ Trap 2: Variable Volume Tank Denominators:** When inflow rate $\neq$ outflow rate, volume is dynamic: $V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$. Forgetting this produces an incorrect constant denominator.